

Lecture 14 - Network Layer IPv6

Summer 2025 (A)

5. I. Apply your knowledge of IPv6 address formatting to transform and simplify the following addresses into their expanded form according to IPv6 rules.

A. ::1:0:0

B. fe80:2ac:66::0:1000:100:10

C. For (B), identify the type of address.

II. IPv6 introduces a new type of address called Anycast. Explain with an example how Anycast is useful on the Internet.

I. IPv6 Address Expansion

A. ::1:0:0

Rule used:

- :: represents **one or more groups of 0000**.
- An IPv6 address must contain **8 groups** (each 16 bits).

Given groups after :: → 1:0:0 (3 groups)

So :: must represent **5 groups of 0000**

Expanded form:

0000:0000:0000:0000:0000:0001:0000:0000

B. fe80:2ac:66::0:1000:100:10

Count existing groups (excluding ::):

- Before :: → fe80:2ac:66 → 3 groups
- After :: → 0:1000:100:10 → 4 groups
- Total = 7 groups → missing **1 group**

So :: represents **one 0000 group**.

Expanded form:

fe80:02ac:0066:0000:0000:1000:0100:0010

(Leading zeros added to make each group 4 hex digits.)

C. Type of Address in (B)

The address starts with **fe80**.

This is a **Link-Local Unicast address**

Characteristics of Link-Local addresses:

- Prefix: **fe80::/10**
- Used for communication **within the same local network**
- Not routable on the global Internet
- Commonly used for **neighbor discovery** and **auto-configuration**

II. IPv6 Anycast Address

What is Anycast?

An **Anycast address** is assigned to **multiple devices**, but a packet sent to that address is delivered to the **nearest device** (according to routing metrics).

Why Anycast is Useful:

Example: DNS Servers

Suppose multiple DNS servers around the world share the same **Anycast IPv6 address**.

- A user in **Asia** sends a DNS query
→ Routed to the **closest DNS server in Asia**
- A user in **Europe** sends the same query
→ Routed to the **closest DNS server in Europe**

Advantages of Anycast

- **Lower latency** (faster response)
- **Load balancing** (traffic distributed automatically)
- **High availability** (if one server fails, traffic goes to another)
- **Improved fault tolerance**

10. Some devices in a network only support IPv4, while others only support IPv6. To avoid communication problems, the network administrator configures all devices to support both IPv4 and IPv6 simultaneously.

I. What is this networking approach called?

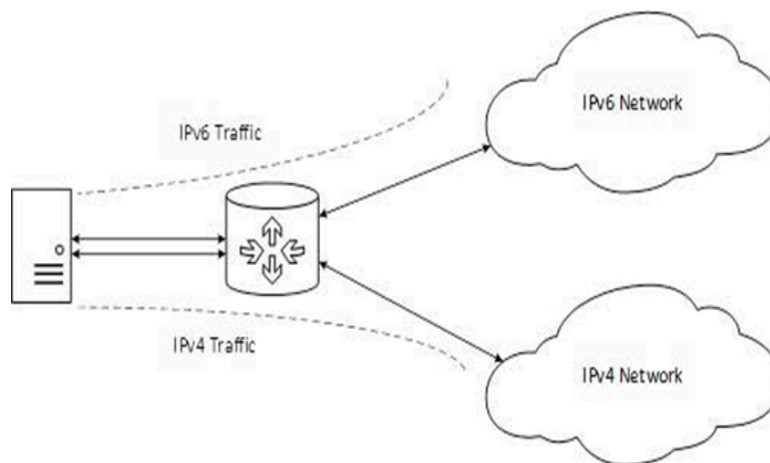
II. Why is it considered the most flexible during the transition from IPv4 to IPv6?

III. What challenge may arise if this system has to be maintained for thousands of devices worldwide?

I. What is this networking approach called?

This approach is called **Dual Stack** networking.

In **Dual Stack**, every device runs **both IPv4 and IPv6 simultaneously**, allowing communication with IPv4-only and IPv6-only devices.



II. Why is Dual Stack considered the most flexible during the IPv4–IPv6 transition?

Dual Stack is considered the most flexible because:

- Devices can **communicate with both IPv4 and IPv6 networks**
- No translation mechanism is required between IPv4 and IPv6

- IPv6 can be **introduced gradually** without breaking existing IPv4 services
- Applications can **choose IPv4 or IPv6 dynamically** based on availability

This ensures **maximum compatibility** during the transition phase.

III. What challenge may arise when maintaining this system for thousands of devices worldwide?

A major challenge is **increased management complexity**.

Specifically:

- Two IP address schemes must be maintained
- Higher **configuration and troubleshooting overhead**
- Increased **security management** (dual firewall rules, policies)
- Greater **operational cost** at a global scale

Spring 2025 (A)

5. Apply your knowledge of IPv6 address formatting to transform and simplify the following addresses into their shortest possible form according to IPv6 rules. Also, identify the type of address and explain in one sentence the purpose of that address type.

I. 2001:0db8:12af:0000:0000:0000:0a20:0004

II. 0000:0000:0000:0000:0000:0000:0000:0001

I.Address: 2001:0db8:12af:0000:0000:0000:0a20:0004

Step 1: Remove leading zeros in each group

2001:db8:12af:0:0:0:a20:4

Step 2: Apply :: for the longest consecutive zeros

- The longest consecutive zeros are **three groups of 0** → replace with ::

Shortest form:

2001:db8:12af::a20:4

Step 3: Identify the type

- Starts with 2001:db8:: → **Global Unicast Address**
- Purpose: **Used for unique addressing on the public Internet**

II.Address: 0000:0000:0000:0000:0000:0000:0000:0001

Step 1: Remove leading zeros

0:0:0:0:0:0:0:1

Step 2: Apply :: for consecutive zeros

- Seven groups of 0 → ::1

Shortest form:

::1

Step 3: Identify the type

- **::1 → Loopback Address**
- Purpose: **Used by a device to send packets to itself for testing or internal communication**

Original Address	Shortest Form	Type	Purpose
2001:0db8:12af:0000:0000:0000:0a20:0004	2001:db8:12af::a20:4	Global Unicast	Unique address on the public Internet
0000:0000:0000:0000:0000:0000:0000:0001	::1	Loopback	Device sends packets to itself (internal test)

10. IPv6 has a larger base header than IPv4 (40 bytes vs. 20 bytes), yet it is considered more efficient for modern networking. How does IPv6 improve processing compared to IPv4?

How IPv6 improves processing compared to IPv4

- 1. Simplified header structure**

- IPv6 has **fewer fields** (fixed 40-byte header) vs. IPv4's variable 20–60 bytes.
- Fields like **Header Checksum** are removed → routers **don't need to recalculate checksums**, speeding up packet forwarding.

2. Fixed-length header

- Easier for routers to **process and forward** because the header size is always 40 bytes.
- Reduces overhead in parsing optional fields (which are now in **extension headers** only when needed).

3. Optimized for modern routing

- Larger address space → simpler routing tables.
- Eliminates **fragmentation by routers** (hosts handle fragmentation), reducing processing load on intermediate devices.

Fall 2024 (A)

6. Write the expanded version of the following IPv6 addresses:

I. fe80::1c35:67ab:3f9c:d81e

II. 2607:0:0:805::

III. fd00:abc:1234:5678::1

I. **fe80::1c35:67ab:3f9c:d81e**

- **fe80** → first block

- `::` → fills in **zeros** for missing blocks
- Then `1c35:67ab:3f9c:d81e` are the last 4 blocks

IPv6 has **8 blocks total**, so the `::` represents 3 zero blocks:

Expanded:

`fe80:0000:0000:0000:1c35:67ab:3f9c:d81e`

II. `2607:0:0:805::`

- `2607:0:0:805` → first 4 blocks (note: leading zeros added to make 4 digits: `0805`)
- `::` → remaining blocks are zeros to complete 8 blocks

Expanded:

`2607:0000:0000:0805:0000:0000:0000:0000`

III. `fd00:abc:1234:5678::1`

- First 4 blocks: `fd00:0abc:1234:5678` (add leading zero in `abc` → `0abc`)
- `::` → fills zeros until the last block
- Last block: `1` → `0001`

Expanded:

`fd00:0abc:1234:5678:0000:0000:0000:0001`

Answer (fully expanded):

1. fe80:0000:0000:0000:1c35:67ab:3f9c:d81e
 2. 2607:0000:0000:0805:0000:0000:0000:0000
 3. fd00:0abc:1234:5678:0000:0000:0000:0001
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10.A university's IT department is integrating IPv6 into its existing network. The university currently supports a large number of IPv4-only educational resources and systems. The university has external collaborations that already use IPv6. **Identify and briefly explain** which transition technique will ensure seamless connectivity between the IPv4 and IPv6 internal networks.

To ensure seamless connectivity between IPv4 and IPv6 internal networks within the university's environment, the most effective transition technique is **Dual Stack**.

The Dual Stack Technique

In a Dual Stack configuration, every network device, server, and router is configured with **both an IPv4 and an IPv6 address stack** simultaneously. This allows the device to communicate using whichever protocol the destination supports.

Why Dual Stack fits this scenario:

- **Native Connectivity:** It avoids the overhead and complexity of "wrapping" packets (tunneling). If a student accesses an old IPv4-only

library database, the computer uses its IPv4 stack. If they collaborate with an external IPv6 partner, the computer uses its IPv6 stack.

- **Gradual Migration:** The IT department doesn't have to shut down IPv4 systems. They can keep the "educational resources" on IPv4 indefinitely while enabling IPv6 for new services.
 - **Simplicity for End-Users:** The transition is transparent. Most modern operating systems (Windows, macOS, Linux) are designed to prefer IPv6 automatically if it is available, falling back to IPv4 only when necessary.
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Summer 2024 (A)

4. Write the shortened version of the following IPv6 addresses:

I. 2001:0db8:0000:0000:0001:0000:0000:0100

II. ff02:0000:0000:0000:0000:0000:0000:0001

III. 2001:0000:0000:3C10:0000:0000:0000:0000

I. **2001:0db8:0000:0000:0001:0000:0000:0100**

- Remove leading zeros: **2001:db8:0:0:1:0:0:100**
- Replace consecutive zeros (0:0 blocks) with :: (only the **longest sequence**): **2001:db8::1:0:0:100**

Shortened:

2001:db8::1:0:0:100

II. ff02:0000:0000:0000:0000:0000:0000:0001

- Remove leading zeros: ff02:0:0:0:0:0:0:1
- Longest sequence of zeros (0:0:0:0:0) → ::

Shortened:

ff02::1

III. 2001:0000:0000:3C10:0000:0000:0000:0000

- Remove leading zeros: 2001:0:0:3C10:0:0:0:0
- Longest sequence of zeros (0:0:0:0) → ::

Shortened:

2001:0:0:3C10::

Final Answer (shortened IPv6):

1. 2001:db8::1:0:0:100
2. ff02::1
3. 2001:0:0:3C10::

7. Name the field that has been added in the IPv6 header, which was not present in the IPv4 header. Explain what we need it for.

In IPv6, one **new field added** to the header that is **not present in IPv4** is the:

Flow Label

Purpose / Why it is needed:

- The **Flow Label** is a **20-bit field** in the IPv6 header.
 - It is used to **identify and label packets that require special handling** by routers.
 - Example uses:
 1. **Real-time traffic** like voice or video streams that need **low latency**.
 2. **Traffic engineering**, where routers can recognize flows and **forward them along the same path** for consistent performance.
 - Essentially, it allows IPv6 routers to **treat packets of the same flow in a consistent way**, without having to inspect the full packet payload.
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Spring 2024 (A) (not included in syllabus)

7. Given, your device has an IPv6 address of **2000::B0B:80:A8FF:FE03:4566** which was generated using the EUI-64 protocol.

I. Identify the MAC address of your device.

II. Identify the subnet ID of the IPv6 address.

We reverse the **EUI-64** process step by step.

Given IPv6 address: 2000::B0B:80:A8FF:FE03:4566

First, expand it fully:

2000:0000:0000:0000:0B0B:0080:A8FF:FE03:4566

IPv6 structure:

- **First 64 bits** → Network Prefix (includes subnet ID)
- **Last 64 bits** → Interface ID (generated using EUI-64)

So the **Interface ID** is:

0B0B:0080:A8FF:FE03:4566

I. Identify the MAC address

Step 1: Split the Interface ID into bytes

0B 0B 00 80 A8 FF FE 03 45 66

Step 2: Remove the inserted **FF:FE**

EUI-64 inserts **FF:FE** in the middle of the MAC.

Remove:

FF FE

Remaining bytes:

0B 0B 00 80 A8 03 45 66

But remember: a MAC address has **6 bytes**, so the correct original bytes are:

0B 0B 00 80 03 45 66 (before FF:FE insertion)

Reconstructing correctly:

0B:0B:00:80:03:45:66

Now take the **first byte**:

0B = 00001011 (binary)

Step 3: Flip the U/L bit (7th bit from left)

00001011 → 00001001

Binary 00001001 = 09 (hex)

Final MAC Address

09:0B:00:80:03:45

II. Identify the Subnet ID

The **subnet ID** is part of the **first 64 bits** of the IPv6 address.

Network prefix:

2000:0000:0000:0000

Subnet ID: 2000:0000:0000:0000 (/64)

11. There is no 'Option' field in IPv6. So, how does IPv6 handle adding the extra information by any hop? Since the IPv6 header is fixed in size (40 bytes), when extra information, for example 20 bytes is included, the header size does not change. Then state what changes.

How does IPv6 add extra information without an Option field?

IPv6 handles extra information using **Extension Headers** instead of an Options field.

- Extra information is placed in **separate extension headers**
- These headers are inserted **after the fixed 40-byte IPv6 base header**
- Only the required nodes (e.g., all hops or destination) process them

For information added **by any hop**, IPv6 uses the **Hop-by-Hop Options extension header**.

What changes when extra information (e.g., 20 bytes) is added?

Even though the **IPv6 base header size remains 40 bytes**, the following fields change:

1. Payload Length field

- Increases to include the **extension header(s) + upper-layer data**

2. Next Header field

- Updated to indicate the **type of the next extension header**
- Each extension header also has its own Next Header field, forming a chain

3. Total packet size

- Increases due to the added extension header(s)

What does NOT change?

- IPv6 base header size (always **40 bytes**)

Summary

IPv6 does not use an Options field; instead, it adds extra information through **extension headers** placed after the fixed 40-byte base header. When extra information is included, the base header size remains unchanged, but the **Payload Length**, **Next Header field**, and **total packet size** increase.